Linear Algebra II (2026 Fall)

Studying Contents :

Chapter 6. Inner Product Spaces

- Section 6.3 : The Adjoint of a Linear Operator

◎ Introduction to Section 6.3: The Adjoint of a Linear Operator

V : an n-dimensional inner product space over F, $T: V \to V$: a linear operator.

In Chapter 5, we studied when T is diagonalizable

– that is, when V has a basis consisting of eigenvectors of T.

If $\beta = \{v_1, \ldots, v_n\}$ is such a basis, with $T(v_i) = \lambda_i v_i$, then

$$T(a_1 v_1 + \cdots + a_n v_n) = a_1 \lambda_1 v_1 + \cdots + a_n \lambda_n v_n.$$

Thus, relative to this basis, T acts simply by multiplying each coordinate by the corresponding eigenvalue.

In Chapter 6, we have an additional structure: an inner product.

This allows us to discuss lengths and orthogonality, and it leads to a stronger question:

When does V have an orthonormal basis consisting of eigenvectors of T?

Why is this useful? Suppose that $\beta = \{v_1, \ldots, v_n\}$ is an orthonormal basis of eigenvectors.

For $x = a_1 v_1 + \cdots + a_n v_n$, we have not only

$$T(a_1 v_1 + \cdots + a_n v_n) = a_1 \lambda_1 v_1 + \cdots + a_n \lambda_n v_n$$

but also

$$\|x\|^2 = |a_1|^2 + \cdots + |a_n|^2 \text{ and } \|T(x)\|^2 = |a_1|^2 |\lambda_1|^2 + \cdots + |a_n|^2 |\lambda_n|^2$$

Because the basis vectors are orthonormal, there are no cross terms.

Comparing these two expressions tells us directly how T changes the length of a vector.

In particular, the magnitude $|\lambda_i|$ describes the change in length along the i-th eigenvector direction.

An orthonormal eigenbasis therefore gives us

a simple description of both the action of T and its effect on lengths.

To determine when such a basis exists, we need

a way to describe how T interacts with the inner product.

This is the purpose of the adjoint operator.

Today, we will construct a linear operator $T^* : V \to V$, called the adjoint of T,

characterized by the identity $\langle T(x), y \rangle = \langle x, T^*(y) \rangle$ for all $x, y \in V$.

In other words, the adjoint allows us

to transfer an operator from one side of an inner product to the other.

This construction will lead to the following characterizations in the next section.

Theorem A : If $F = \mathbb{R}$, then

V has an orthonormal basis of eigenvectors of T if and only if $T = T^*$.

Theorem B : If $F = \mathbb{C}$, then

V has an orthonormal basis of eigenvectors of T if and only if $T T^* = T^* T$.

Our goal in this section is to understand the adjoint and its basic properties,

so that we can use these conditions to study orthonormal eigenbases.

Section 6.3. The adjoint of a linear operator

Recall : (1) For $A \in M_{m\times n}(\mathbb{C})$, we define the conjugate transpose (or adjoint) of A to be

the $n\times m$ matrix A^* such that $(A^*)_{ij} = \overline{A_{ji}}$ for all i,j.

(2) For $\alpha \in M_{1\times 1}(\mathbb{C}) = \mathbb{C}$, we have

$$\alpha^* = \overline{\alpha}.$$

Let $T : V \to V$ be a linear operator on an inner product space V.

The main goal of this section is to define

a related linear operator on V called the adjoint of T,

whose matrix representation with respect to any orthonormal basis β for V is $[T]_\beta^*$.

◎ Linear functions on an inner product space

Let V be an inner product space, and let $y \in V$. Then the function

$$g : V \to F, \quad x \mapsto \langle x, y \rangle,$$

is always a linear transformation.

Now, suppose that V is of dimension n and let $\beta = \{v_1, v_2, \ldots, v_n\}$ be an orthonormal basis for V.

Also write $y = b_1 v_1 + b_2 v_2 + \cdots + b_n v_n$ and $x = a_1 v_1 + a_2 v_2 + \cdots + a_n v_n$.

Then

$$g(x) = \langle x, y \rangle = \langle a_1 v_1 + a_2 v_2 + \cdots + a_n v_n, b_1 v_1 + b_2 v_2 + \cdots + b_n v_n \rangle = a_1 \overline{b_1} + a_2 \overline{b_2} + \cdots + a_n \overline{b_n}.$$

An interesting fact is that

every linear transformation from V into F is of this form

by the following Theorem 6.8.

Theorem 6.8. : Let V be an inner product space of dimension n over F.

If $g : V \to F$ is a linear transformation, then there exists a unique vector $y \in V$ such that

$$g(x) = \langle x, y \rangle \text{ for all } x \in V.$$

Proof : (Existence of $y \in V$) Let $\beta = \{v_1, v_2, \ldots, v_n\}$ be an orthonormal basis for V.

Also, let $b_i = \overline{g(v_i)}$ for all $i = 1, 2, \ldots, n$.

For $x = a_1 v_1 + a_2 v_2 + \cdots + a_n v_n \in V$, we have

$$g(x) = g(a_1 v_1 + a_2 v_2 + \cdots + a_n v_n) = a_1 g(v_1) + a_2 g(v_2) + \cdots + a_n g(v_n) = a_1 \overline{b_1} + a_2 \overline{b_2} + \cdots + a_n \overline{b_n}.$$

Thus, if $y = b_1 v_1 + b_2 v_2 + \cdots + b_n v_n$ then $g(x) = \langle x, y \rangle$ for all $x \in V$.

This proves that

$$g : V \to F \text{ and } \langle -, y \rangle : V \to F \text{ are the same linear functions on } V.$$

(Uniqueness of $y \in V$) Suppose that there exists a vector $z \in V$ such that

$$\langle x, y \rangle = \langle x, z \rangle \text{ for all } x \in V.$$

Then we have $\langle x, y \rangle - \langle x, z \rangle = \langle x, y - z \rangle$ is equal to zero for all $x \in V$.

In particular, $\|y - z\|^2 = \langle y - z, y - z \rangle = 0$ and hence $y - z = O_V$. (See Theorem 6.1.(e).)

This completes the proof that $y = z$. ■

Example 1. : Define $g : \mathbb{R}^2 \to \mathbb{R}$ by $g(a_1, a_2) = 2a_1 + a_2$;

Theorem 6.8	Example 1
$\beta = \{v_1, v_2, \ldots, v_n\}$: an orthonormal basis for V $b_i = \overline{g(v_i)}$ for all $i = 1, 2, \ldots, n$. $y = b_1 v_1 + b_2 v_2 + \cdots + b_n v_n$	Clearly, g is a linear transformation. Let $\beta = \{e_1, e_2\}$, and let $y = g(e_1)e_1 + g(e_2)e_2 = 2e_1 + e_2 = (2,1)$. Then $g(a_1, a_2) = 2a_1 + a_2 = \langle (a_1, a_2), (2,1) \rangle$.

◎ The adjoint operator

Let $T : V \to V$ be a linear operator on an n-dimensional inner product space V.

Also, let $\beta = \{v_1, v_2, \ldots, v_n\}$ be an orthonormal basis for V.

Remark 1. : For each $y \in V$, we will find a vector $T^*(y) \in V$ such that

$$\langle T(x), y \rangle = \langle x, T^*(y) \rangle \text{ for all } x \in V.$$

To this aim, note that the following map is a linear function:

$$g_{T,y} : V \to F, \quad x \mapsto \langle T(x), y \rangle$$

Indeed, we have

$$\begin{aligned} g_{T,y}(c_1x_1 + c_2x_2) &= \langle T(c_1x_1 + c_2x_2), y \rangle \\ &= c_1\langle T(x_1), y \rangle + c_2\langle T(x_2), y \rangle = c_1 g_{T,y}(x_1) + c_2 g_{T,y}(x_2). \end{aligned}$$

Now, Theorem 6.8 says that there exists a unique vector $z \in V$ such that

$$g_{T,y}(x) = \langle T(x), y \rangle \text{ is equal to } \langle x, z \rangle \text{ for all } x \in V.$$

More precisely,

$$z = b_1v_1 + b_2v_2 + \cdots + b_nv_n \text{ where } b_i = \overline{g_{T,y}(v_i)} = \overline{\langle T(v_i), y \rangle} \text{ for } 1 \le i \le n.$$

Theorem 6.9. : Let $T : V \to V$ be a linear operator on an n-dimensional inner product space V.

Then there exists a unique function $T^* : V \to V$ such that

$$\langle T(x), y \rangle = \langle x, T^*(y) \rangle \text{ for all } x, y \in V.$$

Furthermore, $T^* : V \to V$ is a linear transformation.

Remark 2. : The linear operator T^* in Theorem 6.9 is called the adjoint of the operator T.

The symbol T^* is read "T star."

Proof of Existence of T^* : By Remark 1, for each $y \in V$ there exists a unique $z \in V$ where

$$\langle T(x), y \rangle = \langle x, z \rangle \text{ for all } x \in V.$$

More precisely, z depends only on T, $\beta = \{v_1, v_2, \ldots, v_n\}$ and y since

$$z = \overline{\langle T(v_1), y \rangle} v_1 + \overline{\langle T(v_2), y \rangle} v_2 + \cdots + \overline{\langle T(v_n), y \rangle} v_n.$$

Thus the map $T^* : V \to V$ defined as

$$T^*(y) := \overline{\langle T(v_1), y \rangle} v_1 + \overline{\langle T(v_2), y \rangle} v_2 + \cdots + \overline{\langle T(v_n), y \rangle} v_n$$

satisfies the desired condition that $\langle T(x), y \rangle = \langle x, T^*(y) \rangle$ for all $x, y \in V$.

Proof of Uniqueness of T^* : Let $S : V \to V$ be a function such that

$$\langle T(x), y \rangle = \langle x, S(y) \rangle \text{ for all } x, y \in V.$$

Then, for each $y \in V$ we have

$$\langle T(x), y \rangle = \langle x, S(y) \rangle = \langle x, T^*(y) \rangle \text{ for all } x \in V$$

and hence

$$\langle x, S(y) - T^*(y) \rangle = 0 \text{ for all } x \in V.$$

This implies that

$$\|S(y) - T^*(y)\|^2 = \langle S(y) - T^*(y), S(y) - T^*(y) \rangle = 0$$

and so $S(y) - T^*(y) = O_V$.

This shows that

$$S(y) = T^*(y) \text{ for all } y \in V \text{ and so } S = T^*.$$

Proof of Linearity of T^* : We know already that $T^* : V \to V$ is defined by

$$T^*(y) := \overline{\langle T(v_1), y\rangle}\, v_1 + \overline{\langle T(v_2), y\rangle}\, v_2 + \cdots + \overline{\langle T(v_n), y\rangle}\, v_n.$$

Thus we have

$$\begin{aligned} T^*(c_1y_1 + c_2y_2) &= \overline{\langle T(v_1), c_1y_1 + c_2y_2\rangle}\, v_1 + \overline{\langle T(v_2), c_1y_1 + c_2y_2\rangle}\, v_2 + \cdots + \overline{\langle T(v_n), c_1y_1 + c_2y_2\rangle}\, v_n \\ &= \sum_{i=1}^{n} \left(c_1 \overline{\langle T(v_i), y_1\rangle} + c_2 \overline{\langle T(v_i), y_2\rangle} \right) v_i \end{aligned}$$

since for each $1 \le i \le n$, it holds that

$$\overline{\langle T(v_i), c_1y_1 + c_2y_2\rangle} = \overline{\overline{c_1}\langle T(v_i), y_1\rangle + \overline{c_2}\langle T(v_i), y_2\rangle} = c_1\overline{\langle T(v_i), y_1\rangle} + c_2\overline{\langle T(v_i), y_2\rangle}.$$

Thus it follows that

$$\begin{aligned} T^*(c_1y_1 + c_2y_2) &= \sum_{i=1}^{n} \left(c_1 \overline{\langle T(v_i), y_1\rangle} + c_2 \overline{\langle T(v_i), y_2\rangle} \right) v_i \\ &= c_1 \left(\sum_{i=1}^{n} \overline{\langle T(v_i), y_1\rangle}\, v_i \right) + c_2 \left(\sum_{i=1}^{n} \overline{\langle T(v_i), y_2\rangle}\, v_i \right) \\ &= c_1\, T^*(y_1) + c_2\, T^*(y_2). \end{aligned}$$

■

Remark 3. : Let $T: V \to V$ be a linear operator on an n-dimensional inner product space V.

Then

$$\langle T(x), y \rangle = \langle x, T^*(y) \rangle \text{ for all } x, y \in V.$$

Note that we also have

$$\langle x, T(y) \rangle = \overline{\langle T(y), x \rangle} = \overline{\langle y, T^*(x) \rangle} = \langle T^*(x), y \rangle.$$

The equations

$$\langle T(x), y \rangle = \langle x, T^*(y) \rangle \text{ and } \langle x, T(y) \rangle = \langle T^*(x), y \rangle$$

can be viewed symbolically as

adding a * to T when shifting its position inside the inner product symbol.

Remark 4. : For an infinite-dimensional inner product space,

the adjoint of a linear operator T may be defined to be

the function $T^* : V \to V$ such that $\langle T(x), y \rangle = \langle x, T^*(y) \rangle$ for all $x, y \in V$,

provided it exists.

(1) The uniqueness and linearity of $T^* : V \to V$ follow as before.

(2) The existence of the adjoint is not guaranteed in general (Exercise 24).

(3) The hypothesis of finite dimensionality in the proof of Theorem 6.8 is used to

construct the element $y = \sum_{i=1}^{n} \overline{g(v_i)}\, v_i$ such that $g(x) = \langle x, y \rangle$ for all $x \in V$.

Here, if V has infinite dimension, then

y would be an infinite sum, which means that it may not be well-defined.

Exercise #23. in Section 6.2 : Let $V = \{ (a_1, a_2, \ldots) \mid a_i \in F,\ a_i \neq 0 \text{ for only finitely many } i \}$.

Thus V is an infinite dimensional vector space over F

and $\beta = \{ e_1, e_2, \ldots, e_n, \ldots \}$ is a basis for V.

For $\sigma = (a_1, a_2, \ldots)$ and $\tau = (b_1, b_2, \ldots)$, we define $\langle \sigma, \tau \rangle = \sum_{i=1}^{\infty} a_i \overline{b_i}$.

(a) $\langle \cdot, \cdot \rangle$ is an inner product on V.

(b) $\beta = \{ e_1, e_2, \ldots, e_n, \ldots \}$ is an orthonormal basis for V.

(c) For each $n \geq 2$, let $\sigma_n = e_1 + e_n$. Also, let $W = \text{span}(\{\sigma_n \mid n \geq 2\})$.

Then (i) $e_1 \notin W$ and so $W \neq V$; and (ii) $W^{\perp} = \{0_V\}$.

Exercise #24. : Define $T : V \to V$ by

$$(a_1, a_2, \ldots) \mapsto \left(\sum_{i=1}^{\infty} a_i, \sum_{i=2}^{\infty} a_i, \ \ldots \right).$$

(a) T is a linear operator on V.

(b) $T(e_n) = e_1 + \cdots + e_n$ for every $n \geq 1$.

(c) T has no adjoint.

(c) T has no adjoint.

Solution : By way of contradiction, suppose that T^* exists.

Write $T^*(e_1) = a_1e_1 + a_2e_2 + \cdots$.

Then we have

$$\langle T(e_n), e_1 \rangle = \langle e_1 + \cdots + e_n, e_1 \rangle = 1$$

$$\| \qquad\qquad\qquad\

Theorem 6.10. : Let $T: V \to V$ be a linear operator on an n-dimensional inner product space V. Also, let $\beta = \{v_1, v_2, \ldots, v_n\}$ be an orthonormal basis for V. Then

$$[T^*]_\beta = [T]_\beta^*.$$

Proof : Let $A = [T]_\beta$ and $B = [T^*]_\beta$. By Corollary to Theorem 6.5, it holds that

$$B_{ij} = \langle T^*(v_j), v_i \rangle \text{ for all } 1 \le i, j \le n.$$

Thus we have

$$B_{ij} = \langle T^*(v_j), v_i \rangle = \langle v_j, T(v_i) \rangle = \overline{\langle T(v_i), v_j \rangle} = \overline{A_{ji}} = (A^*)_{ij}.$$

This proves that $B = A^*$. ■

Corollary : Let $A \in M_{n \times n}(F)$. Then $(L_A)^* = L_{A^*}$.

Proof : Let $\beta = \{e_1, e_2, \ldots, e_n\}$ be the standard ordered basis for F^n. Then

$$[L_A]_\beta = A \text{ and } [L_{A^*}]_\beta = A^*.$$

Therefore we have $[L_A^*]_\beta = [L_A]_\beta^* = A^* = [L_{A^*}]_\beta$ and hence $(L_A)^* = L_{A^*}$. ■

Example 2. : Let $T : \mathbb{C}^2 \to \mathbb{C}^2$ be the linear operator on $\mathbb{C}^2$ defined by

$$T(a_1, a_2) = (2ia_1 + 3a_2, a_1 - a_2).$$

If $\beta = \{(1,0), (0,1)\}$ is the standard basis for $\mathbb{C}^2$, then

$$[T]_\beta = \begin{pmatrix} 2i & 3 \\ 1 & -1 \end{pmatrix}.$$

So, it holds that

$$[T^*]_\beta = [T]_\beta^* = \begin{pmatrix} -2i & 1 \\ 3 & -1 \end{pmatrix}.$$

Hence it follows that

$$T^*(a_1, a_2) = (-2ia_1 + a_2, 3a_1 - a_2).$$

Theorem 6.11. : Let T and U be linear operators on an inner product space V. Then

(1) $(T+U)^* = T^* + U^*$;

(2) $(cT)^* = \bar{c}\,T^*$ for any $c \in F$;

(3) $(TU)^* = U^*T^*$;

(4) $T^{**} = T$;

(5) $I^* = I$.

Proof : (1) Let $x, y \in V$. Then

$$\begin{aligned}\langle x, (\mathsf{T}+\mathsf{U})^*(y)\rangle &= \langle (\mathsf{T}+\mathsf{U})(x), y\rangle = \langle \mathsf{T}(x)+\mathsf{U}(x), y\rangle \\ &= \langle \mathsf{T}(x), y\rangle + \langle \mathsf{U}(x), y\rangle = \langle x, \mathsf{T}^*(y)\rangle + \langle x, \mathsf{U}^*(y)\rangle \\ &= \langle x, \mathsf{T}^*(y)+\mathsf{U}^*(y)\rangle = \langle x, (\mathsf{T}^*+\mathsf{U}^*)(y)\rangle\,,\end{aligned}$$

and hence $\langle x, (T+U)^*(y) - (T^*(y)+U^*(y))\rangle = 0$ for all $x \in V$.

This implies that $(T+U)^*(y) - (T^*(y)+U^*(y)) = O_V$ and so

$$(T+U)^*(y) = (T^*(y)+U^*(y)) \text{ for all } y \in V.$$

This proves the desired equality $(T+U)^* = T^* + U^*$.

Theorem 6.11. : Let T and U be linear operators on an inner product space V. Then

(1) $(T+U)^* = T^* + U^*$;

(2) $(cT)^* = \bar{c}\,T^*$ for any $c \in F$;

(3) $(TU)^* = U^*T^*$;

(4) $T^{**} = T$;

(5) $I^* = I$.

Proof : (2) Let $x, y \in V$. Then

$$\begin{aligned}\langle x, (cT)^*(y)\rangle &= \langle (cT)(x), y\rangle = \langle cT(x), y\rangle \\ &= \langle T(x), \bar{c}y\rangle = \langle x, T^*(\bar{c}y)\rangle \\ &= \langle x, \bar{c}\,T^*(y)\rangle.\end{aligned}$$

Thus $\langle x, (cT)^*(y) - T^*(\bar{c}y)\rangle = 0$ and so $(cT)^*(y) - \bar{c}\,T^*(y) = O_V$.

Again, this implies that $(cT)^* = \bar{c}\,T^*$.

Theorem 6.11. : Let T and U be linear operators on an inner product space V. Then

(1) $(T+U)^* = T^* + U^*$;

(2) $(cT)^* = \bar{c}\, T^*$ for any $c \in F$;

(3) $(TU)^* = U^* T^*$;

(4) $T^{**} = T$;

(5) $I^* = I$.

Proof : (3) Let $x, y \in V$. Then

$$\begin{aligned} \langle x, (TU)^*(y) \rangle &= \langle (TU)(x), y \rangle = \langle T(U(x)), y \rangle \\ &= \langle U(x), T^*(y) \rangle = \langle x, U^*(T^*(y)) \rangle \\ &= \langle x, (U^* T^*)(y) \rangle. \end{aligned}$$

So, as in the previous cases, this implies that $(TU)^* = U^* T^*$.

(4) From $\langle T(x), y \rangle = \langle x, T^*(y) \rangle = \langle T^{**}(x), y \rangle$, it follows that $T^{**} = T$.

(5) From $\langle x, I(y) \rangle = \langle x, y \rangle = \langle I(x), y \rangle = \langle x, I^*(y) \rangle$, it follows that $I^* = I$. ■

Remark 5. : The same proof works in the infinite-dimensional case, provided that the existence of T^* and U^* is assumed.

More generally, many of the theorems we prove about adjoints do not depend on V being finite-dimensional.

Thus, unless stated otherwise, for the remainder of this chapter we adopt the convention that a reference to the adjoint of a linear operator on an infinite dimensional inner product space assumes its existence.

Corollary : Let A and B be $n \times n$ matrices. Then

(1) $(A+B)^* = A^* + B^*$;

(2) $(cA)^* = \bar{c}A^*$ for all $c \in F$;

(3) $(AB)^* = B^*A^*$;

(4) $A^{**} = A$;

(5) $I^* = I$.

Proof : (3) We have

$$L_{(AB)^*} = (L_{AB})^* = (L_A L_B)^* = L_B^* L_A^* = L_{B^*} L_{A^*} = L_{B^*A^*},$$

which implies the desired equality

$$(AB)^* = B^*A^*.$$

■

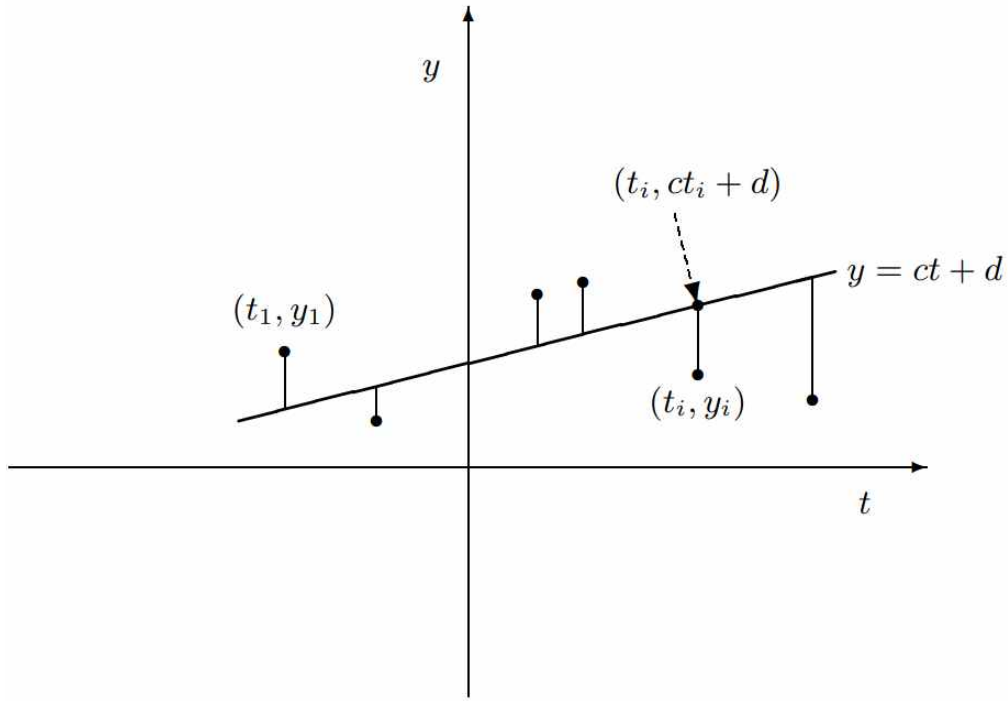
◎ Least Squares Approximation

Problem Introduction :

- The government is measuring unemployment at various times during some period.
- Suppose that the data $(t_1, y_1), (t_2, y_2), \ldots, (t_m, y_m)$ are plotted as points in the plane.
- From this plot, the experimenter feels that there exists an essentially linear relationship between y and t, say

$$y = ct + d,$$

and would like to find the constants c and d so that the line $y = ct + d$ represents the best possible fit to the data collected.



Remark 6. : One such estimate of fit is to calculate the error E that represents the sum of the squares of the vertical distances from the points to the line.

$$E = \sum_{i=1}^{m} (y_i - ct_i - d)^2.$$

- Thus the problem is reduced to finding the constants c and d that minimize E.
- For this reason, the line $y = ct + d$ is called the least squares line.
- If we let

$$A = \begin{pmatrix} t_1 & 1 \\ t_2 & 1 \\ \vdots & \vdots \\ t_m & 1 \end{pmatrix}, \quad x = \begin{pmatrix} c \\ d \end{pmatrix}, \quad \text{and} \quad y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{pmatrix},$$

then it follows that $y - Ax = \begin{pmatrix} y_1 - t_1 c - d \\ y_2 - t_2 c - d \\ \vdots \\ y_m - t_m c - d \end{pmatrix}$ and so $E = \|y - Ax\|^2$.

This leads us to develop a general method for

finding an explicit vector $x_0 \in F^n$ that minimizes $E = \|y - Ax\|^2$.

Problem : Given $A \in M_{m\times n}(F)$ and $y \in F^m$, find $x_0 \in F^n$ such that

$$\|y - Ax_0\| \leq \|y - Ax\| \text{ for all vectors } x \in F^n.$$

※ This method not only allows us to find the linear function that best fits the data, but also, for any positive integer n, the best fit using a polynomial of degree at most n.

Notation : For $x, y \in F^n$, let $\langle x, y\rangle_n$ denote the standard inner product of x and y in F^n.

If we regard x and y as column vectors, then $\langle x, y\rangle_n = y^* x$.

Lemma 1. : Let $A \in M_{m\times n}(F)$, $x \in F^n$ and $y \in F^m$. Then

$$\langle Ax, y\rangle_m = \langle x, A^* y\rangle_n.$$

Proof : By the corollary to Theorem 6.11, we have

$$\langle Ax, y\rangle_m = y^*(Ax) = (y^* A)x = (A^* y)^* x = \langle x, A^* y\rangle_n.$$ ■

Lemma 2. : Let $A \in M_{m\times n}(F)$. Then $\mathrm{rank}(A^{*}A) = \mathrm{rank}(A)$.

Proof : Consider the diagram

$$F^{n} \xrightarrow{A} F^{m} \xrightarrow{A^{*}} F^{n}.$$

Recall that $L_{A} : F^{n} \to F^{m}$ and $\mathrm{rank}(A) = \dim A(F^{n})$.

Also $\mathrm{rank}(A^{*}A) = \dim A^{*}(A(F^{n}))$.

Therefore it is enough to show that $L_{A^{*}} : A(F^{n}) \to F^{n}$ is injective.

Equivalently, we need to show that for $x \in F^{n}$,

$$A^{*}Ax = O \text{ if and only if } Ax = O.$$

Clearly, $Ax = O$ implies $A^{*}Ax = O$.

Conversely, if $A^{*}Ax = O$ then we have

$$0 = \langle A^{*}Ax, x\rangle_{n} = \langle Ax, A^{**}x\rangle = \langle Ax, Ax\rangle_{m} \text{ and so } Ax = O. \qquad \blacksquare$$

Corollary. : Let $A \in M_{m\times n}(F)$ be such that $\mathrm{rank}(A) = n$. Then $A^{*}A$ is invertible.

(♣) Problem : Given $A \in M_{m\times n}(F)$ and $y \in F^m$, find $x_0 \in F^n$ such that

$$\|y - Ax_0\| \leq \|y - Ax\| \text{ for all vectors } x \in F^n.$$

Step 1. : Define $W = \{ Ax \mid x \in F^n \}$. That is, $W = R(L_A)$.

Step 2. : There exists a unique vector in W that is closest to y. (Corollary to Theorem 6.6)

Call this vector Ax_0, where $x_0 \in F^n$.

Step 3. : We have $\|Ax_0 - y\| \leq \|Ax - y\|$ for all $x \in F^n$.

This shows that $x_0 \in F^n$ has the property that $E = \|Ax_0 - y\|$ is minimal, as desired.

Next Problem : Develop a practical method for finding an $x_0 \in F^n$ in Step 2.

- Note that $Ax_0 - y$ is contained in $W^{\perp}$ and so $\langle Ax, Ax_0 - y \rangle_m = 0$ for all $x \in F^n$.
- By Lemma 1, this implies that $\langle x, A^*(Ax_0 - y) \rangle_n = 0$ for all $x \in F^n$.

  In particular, $\langle A^*(Ax_0 - y), A^*(Ax_0 - y) \rangle_n = 0$ and hence $A^*(Ax_0 - y) = O$.
- We need to find a solution $x_0 \in F^n$ to the equation $A^*Ax = A^*y$.
- If, in addition, we assume that $\mathrm{rank}(A) = n$, then

  A^*A is invertible by Lemma 2

  and hence it follows that

$$x_0 = (A^*A)^{-1}(A^*y).$$

Theorem 6.12. : Let $A \in M_{m\times n}(F)$ and $y \in F^m$. Then there exists $x_0 \in F^n$ such that

$$A^*Ax_0 = A^*y \text{ and } \|Ax_0 - y\| \leq \|Ax - y\| \text{ for all } x \in F^n.$$

Furthermore, if $\mathrm{rank}(A) = n$, then $x_0 = (A^*A)^{-1}A^*y$.

To return to our experimenter, let us suppose that the data collected consists of the set

$$\{(t_i, y_i) \mid i = 1,2,3,4\} = \{(1,2), (2,3), (3,5), (4,7)\}.$$

Thus $A = \begin{pmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{pmatrix}$ and $y = \begin{pmatrix} 2 \\ 3 \\ 5 \\ 7 \end{pmatrix}$;

Then $A^* A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{pmatrix} = \begin{pmatrix} 30 & 10 \\ 10 & 4 \end{pmatrix}$ and $(A^* A)^{-1} = \begin{pmatrix} 30 & 10 \\ 10 & 4 \end{pmatrix}^{-1} = \frac{1}{20} \begin{pmatrix} 4 & -10 \\ -10 & 30 \end{pmatrix}$

Therefore

$$\begin{bmatrix} c \\ d \end{bmatrix} = x_0 = (A^* A)^{-1} A^* y = \frac{1}{20} \begin{pmatrix} 4 & -10 \\ -10 & 30 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ 5 \\ 7 \end{pmatrix} = \frac{1}{20} \begin{pmatrix} 4 & -10 \\ -10 & 30 \end{pmatrix} \begin{pmatrix} 51 \\ 17 \end{pmatrix} = \begin{pmatrix} 1.7 \\ 0 \end{pmatrix}.$$

It follows that

(i) the line $y = 1.7\, t$ is the least squares line, and

(ii) the error E is equal to

$$\|Ax_0 - y\|^2 = \|(-0.3, 0.4, 0.1, -0.2)\|^2 = 0.3.$$

Remark 1. : Suppose that $t_1 + t_2 + \cdots + t_m = 0$.

(For example, let $t = \dfrac{t_1 + \cdots + t_m}{n}$ be the average of initially given times.

Then

$$t_1' = t_1 - t, t_2' = t_2 - t, \ldots, t_m' = t_m - t$$

satisfy the condition $t_1' + t_2' + \cdots + t_m' = 0$.)

Now, $A = \begin{pmatrix} t_1 & 1 \\ t_2 & 1 \\ \vdots & \vdots \\ t_m & 1 \end{pmatrix}$ and so

$$A^* A = \begin{pmatrix} t_1 & t_2 \cdots & t_m \\ 1 & 1 \cdots & 1 \end{pmatrix} \begin{pmatrix} t_1 & 1 \\ t_2 & 1 \\ \vdots & \vdots \\ t_m & 1 \end{pmatrix} = \begin{pmatrix} \sum_{i=1}^{m} t_i^2 & 0 \\ 0 & m \end{pmatrix}$$ is a diagonal matrix.

Thus the computation of $x_0 = (A^* A)^{-1} A^* y$ is greatly simplified.

Now, using the same data, we want to find the quadratic polynomial

$$y = ct^2 + dt + e.$$

$$A = \begin{pmatrix} t_1^2 & t_1 & 1 \\ \vdots & \vdots & \vdots \\ t_m^2 & t_m & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 4 & 2 & 1 \\ 9 & 3 & 1 \\ 16 & 4 & 1 \end{pmatrix}, \quad x = \begin{pmatrix} c \\ d \\ e \end{pmatrix}, \quad y = \begin{pmatrix} 2 \\ 3 \\ 5 \\ 7 \end{pmatrix}$$

Our goal is to find $x_0 \in \mathbb{R}^3$ that minimizes

$$E = \|Ax - y\|^2.$$

By Theorem 6.12, it suffices to solve the normal equation

$$A^* A x = A^* y.$$

Since $A^* A = \begin{pmatrix} 1 & 4 & 9 & 16 \\ 1 & 2 & 3 & 4 \\ 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 4 & 2 & 1 \\ 9 & 3 & 1 \\ 16 & 4 & 1 \end{pmatrix} = \begin{pmatrix} 354 & 100 & 30 \\ 100 & 30 & 10 \\ 30 & 10 & 4 \end{pmatrix}$ and $A^* y = \begin{pmatrix} 1 & 4 & 9 & 16 \\ 1 & 2 & 3 & 4 \\ 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ 5 \\ 7 \end{pmatrix} = \begin{pmatrix} 171 \\ 51 \\ 17 \end{pmatrix}$,

the above equation becomes

$$\begin{pmatrix} 354 & 100 & 30 \\ 100 & 30 & 10 \\ 30 & 10 & 4 \end{pmatrix} \begin{pmatrix} c \\ d \\ e \end{pmatrix} = \begin{pmatrix} 171 \\ 51 \\ 17 \end{pmatrix}.$$

Here $\mathrm{rank}(A) = 3$ and hence A^*A is invertible.

Indeed,

$$(A^*A)^{-1}(A^*y) = \begin{pmatrix} 354 & 100 & 30 \\ 100 & 30 & 10 \\ 30 & 10 & 4 \end{pmatrix}^{-1} \begin{pmatrix} 171 \\ 51 \\ 17 \end{pmatrix} = \begin{pmatrix} \frac{1}{4} & -\frac{5}{4} & \frac{5}{4} \\ -\frac{5}{4} & \frac{129}{20} & -\frac{27}{4} \\ \frac{5}{4} & -\frac{27}{4} & \frac{31}{4} \end{pmatrix} \begin{pmatrix} 171 \\ 51 \\ 17 \end{pmatrix} = \begin{pmatrix} \frac{1}{4} \\ \frac{9}{20} \\ \frac{5}{4} \end{pmatrix}.$$

Hence, the least squares quadratic polynomial is $y = \frac{1}{4}t^2 + \frac{9}{20}t + \frac{5}{4}$.

Furthermore, the sum of squared errors is

$$E = \|Ax_0 - y\|^2 = (-0.05)^2 + (0.15)^2 + (-0.15)^2 + (-0.05)^2 = 0.05.$$

Approximation	Least squares polynomial	Squared error E
Linear	$y = 1.7\,t$	0.3
Quadratic	$y = \frac{1}{4}t^2 + \frac{9}{20}t + \frac{5}{4}$	0.05

◎ Minimal Solutions to Systems of Linear Equations

Problem : Let $Ax = b$ be a system of linear equations which is consistent.

When there are more than two solutions (and so infinitely many solutions),

find a solution of minimal norm s in the sense that

$$\|s\| \leq \|u\| \text{ for all other solutions } u.$$

Theorem 6.13. : Let $A \in M_{m\times n}(F)$ and $b \in F^m$ be such that $Ax = b$ is consistent.

Then the following statements are true.

(1) There exists exactly one minimal solution s of $Ax = b$, and $s \in R(L_{A^*})$.

(2) The vector s is the only solution to $Ax = b$ that lies in $R(L_{A^*})$;

That is, if u satisfies $(AA^*)u = b$, then $s = A^*u$.

Lemma A. : Let $A \in M_{m\times n}(F)$ and $b \in F^m$ be such that $Ax = b$ is consistent.

Let x_0 be any solution to $Ax = b$, and

let s be the orthogonal projection of x_0 onto $N(A)^{\perp}$. Then

(1) $As = b$

(2) For any solution v of $Ax = b$, it holds that

$$\|s\| \leq \|v\| \text{ with equality if and only if } v = s.$$

In particular, s is the only solution of $Ax = b$ which is contained in $N(A)^{\perp}$.

Proof : (1) Let $x_0 = s+u$ where $u \in (N(A)^{\perp})^{\perp} = N(A)$.

Then we have $Ax_0 = A(s+u) = As+Au = As = b$.

(2) If v is a solution of $Ax = b$, then $z := v-s \in N(A)$.

In particular, s and z are orthogonal.

Thus we have

$$\|v\|^2 = \|s+z\|^2 = \|s\|^2+\|z\|^2 \geq \|s\|^2$$

with equality if and only if $z = 0$ and hence $v = s$. ■

Lemma B. : Let $A \in M_{m\times n}(F)$. Then $N(A)^{\perp} = R(A^{*})$.

Proof : (Exercise #12) Let $T : V \to V$ be a linear operator on an inner product space V.

Then $R(T^{*})^{\perp} = N(T)$.

($\because$) Let $\beta = \{v_1, v_2, \ldots, v_n\}$ be a basis for V.

Then $R(T^{*}) = \text{span}(T^{*}(v_1), T^{*}(v_2), \ldots, T^{*}(v_n))$.

Thus it holds that

$$\begin{aligned} v \in R(T^{*})^{\perp} &\Leftrightarrow \langle v, T^{*}(v_1)\rangle = \langle v, T^{*}(v_2)\rangle = \cdots = \langle v, T^{*}(v_n)\rangle = 0 \\ &\Leftrightarrow \langle T(v), v_1\rangle = \langle T(v), v_2\rangle = \cdots = \langle T(v), v_n\rangle = 0 \\ &\Leftrightarrow T(v) \in \text{span}(v_1, v_2, \ldots, v_n)^{\perp} = V^{\perp} = \{O_V\} \\ &\Leftrightarrow T(v) = O_V \\ &\Leftrightarrow v \in N(T). \end{aligned}$$

It follows by Exercise #12 that $R(A^{*})^{\perp} = N(A)$, and hence we have $N(A)^{\perp} = R(A^{*})$. ■

Theorem 6.13. : Let $A \in M_{m\times n}(F)$ and $b \in F^m$ be such that $Ax = b$ is consistent.

Then the following statements are true.

(1) There exists exactly one minimal solution s of $Ax = b$, and $s \in R(L_{A^*})$.

(2) The vector s is the only solution to $Ax = b$ that lies in $R(L_{A^*})$;

That is, if u satisfies $(AA^*)u = b$, then $s = A^*u$.

Proof : (1) By Lemma A, there exists a unique s in $N(A)^\perp$ which satisfies $Ax = b$.

Furthermore, solution v of $Ax = b$, it holds that

$$\|s\| \leq \|v\| \text{ with equality if and only if } v = s.$$

By Lemma B, we have $N(A)^\perp = R(A^*)$.

(2) If u satisfies $(AA^*)u = b$, then $A^*u \in R(A^*)$ is a solution of $Ax = b$.

Therefore, $s = A^*u$.

Example 3. : Find the minimal solution to the system

$$\begin{cases} x+2y+\ z= \ 4 \\ x-\ y+2z=-11. \\ x+5y\quad\ = \ 19 \end{cases}$$

Step 1. : Let $A = \begin{pmatrix} 1 & 2 & 1 \\ 1 & -1 & 2 \\ 1 & 5 & 0 \end{pmatrix}$ and $b = \begin{pmatrix} 4 \\ -11 \\ 19 \end{pmatrix}$.

Step 2. : $AA^* = \begin{pmatrix} 1 & 2 & 1 \\ 1 & -1 & 2 \\ 1 & 5 & 0 \end{pmatrix}\begin{pmatrix} 1 & 1 & 1 \\ 2 & -1 & 5 \\ 1 & 2 & 0 \end{pmatrix} = \begin{pmatrix} 6 & 1 & 11 \\ 1 & 6 & -4 \\ 11 & -4 & 26 \end{pmatrix}$

Step 3. : Find some solution u to $AA^*x = b$:

$$\begin{pmatrix} 6 & 1 & 11 \\ 1 & 6 & -4 \\ 11 & -4 & 26 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 4 \\ -11 \\ 19 \end{pmatrix}$$

One solution is $u = \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix}$.

Step 3. : Hence the minimal solution to the given system is

$$s = A^*u = \begin{pmatrix} 1 & 1 & 1 \\ 2 & -1 & 5 \\ 1 & 2 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ 4 \\ -3 \end{pmatrix}.$$